Template: Validation Matrix to MEASURE your concept

What will you test?	How will you test it?	Which stakeholders could be relevant to include in testing?	What will you measure? What data will you collect?	How will you proceed if your test results are true?	How will you proceed if your concept needs modification?
We assume that 50% or above of all Nexø roof top area will be enough to produce enough power.	Data on how many roof tops are viable for solar panels in Nexø should be collected. Calculations on how much energy needs to be produced on the roof top area vs the solar power plant should be made.	Bornholm Municipality; Solar PV engineers	Total rooftop area; percentage of suitable rooftop area	We proceed with our original plan of using	Reduce the role of rooftop PV and emphasize more on a PV park.
We assume that more than 50% of Nexø citizens would like to participate.	We can conduct a survey with residents asking if they would like to have solar panels on their rooftops and whether they would be interrested in renting their roof top out to the project. A layer should be consulted to explore whether the project could rent the city rooftops and how the agreement would could work.	Citizens of Nexø; Bornholm Municipality	Number of responses; Number of citizens willing to participate	Continue with residential rooftop PV and investigate implementation further.	Investigate why citizens are unwilling, consider incentives, or rely more on public/commercial rooftops.
We assume, that the combined systems would be enough to meet both local heat and electric energy demand.	Calculations should be made to find out how much energy this combined system could be produced. Data should be collected on how big the energy and heat demand in the area is, and what the current and future deficit is and will be. Build a simplified energy-system model and compare estimated energy production with local demand.	Bornholms Energi & Forsyning; energy experts/engineers	Electricity production; heat production; electricity demand; heat demand; percentage of demand covered	Continue developing the combined system and optimize its size.	Resize the system, change the combination of technologies, or change the target to covering part of the local demand.
We assume that heat storage at the electric boiler along with one battery at the PV and one combined battery for the roof top PV will be enough.	Data on similar solar power plants should be collected to estimate how much energy will be produced by the solar plants and how much of it can be used immediately and how much will need to be stored. Distance and placement of the batteries would need to be taken into consideration. Here ata from similar projects would need to be collected and evaluated Build a simplified storage model and compare storage capacity with estimated excess energy and demand.	Bornholms Energi & Forsyning; battery/heat-storage experts or suppliers	Required battery capacity; required heat-storage capacity; proposed capacity; excess electricity/heat	Keep the proposed storage structure and develop the design in more detail.	Increase storage capacity, add storage units, or redesign the storage configuration.

Template: Prototyping plan

Day 12
Start

A low-fidelity visual prototype of the proposed energy system.

Which kind of prototype will we do?

What does our prototype need to help us answer.

How could PV park, rooftop PV, battery storage and heat storage be combined in a local energy system to reduce pressure on the electricity grid in Nexø?

Lunch

First, create a visual energy-flow diagram showing the PV park, rooftop PV, battery storage, heat storage, electric boiler, grid and households. Use arrows to show how electricity and heat move between the different parts of the system. The prototype will show how the different technologies could work together as one possible energy solution for Nexø.

How will we build/make the prototype?

Start

Start making the prototype

Create the physical version of the energy-flow diagram. We will focus on the Roof Top PV Park to how the solar panels and the battery will be connected households.

Finish the prototype

Complete the physical prototype, components and connections are included.

Lunch

Make a small test of the prototype

Show the prototype to classmates or people familiar with energy systems. Ask them whether the energy flow is easy to understand, whether the combination of technologies makes sense, and whether any important connections or components are missing. Record their feedback and use it to improve the prototype.

Day 13
end